Plant Stress 7 (2023) 100130

Contents lists available at ScienceDirect

Plant Stress

journal homepage: www.sciencedirect.com/journal/plant-stress

# Physiological and genetic responses of lentil (*Lens culinaris*) under flood stress

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ARTICLE INFO

*Keywords:*
Lentil
qPCR
Volatile
Flood

ABSTRACT

Lentils (*Lens culinaris*) cannot withstand prolonged flooding. This project aimed to study molecular and physiological responses in *L. culinaris* under flood stress to understand this plant's key mechanisms for flood tolerance. Six- to eight-week-old lentil seedlings were flooded in a growth chamber for four days. On the fourth day, molecular and physiological experiments were conducted on the flooded and control lentil plants. In addition, a morphological study of the stems and leaves using a scanning electron microscope and volatile compound studies (serving as potential biomarkers for flood stress) using GC–MS were conducted. Under flooded conditions in most of the lentil plants, the expressions of the following genes were upregulated: *acetyl-CoA carboxylase carboxyltransferase beta subunit (accD), aldolase*, and *petD-cytochrome b6/f complex subunit IV (PetD)*. On the other hand, *chloroplast copper/zinc superoxide dismutase (SD)* and *PBA flash photosystem II D1 precursor (psbA)* constituted two downregulated genes among most of the plants. Additionally, higher levels of absorbance for soluble sugar, surface roughness, and surface wax deposition were observed in lentils under flooded conditions. Higher wax accumulation leads to higher leaf hydrophobicity, which is a flood-related adaptation. On the contrary, chlorophyll content, catalase activity, and volatile emissions were reduced among flooded lentils, probably due to cell death and reduced photosynthesis. Oxidative damage occurs when cells' catalase activity is low, meaning cells are unable to detoxify reactive oxygen species. In conclusion, flooding can cause a notable difference in lentils at anatomical, physiological, and molecular levels.

## 1. Introduction

One of the issues related to climate change is the increased risk of flooding around the world[1]. The rise in floods does not just affect the environment but also causes a considerable amount of damage to farmers around the world. As a result of floods, farmers worldwide lost $21 billion in crops and livestock from 2008 to 2018.[2] Like other crops, lentil (*Lens culinaris*) is a vital food resource worldwide. Many countries use it in dishes like the mshosh in Armenia or dal in India. Comparable to other staple crops, lentils are also a major source of protein, especially for vegetarians. Like other pulses, lentils have high amounts of protein and fiber and are also low in fats.[3] Lentils are also rich in folate, potassium, and iron which are appropriate substitutes for meat[3]. Unfortunately, lentils cannot endure prolonged flooding in areas with heavy rainfall like India.[4] The growth and overall physiology are severely affected by flooding (Nessa et al., 2007). India recently lost 8.5% of total gross cropped areas from floods between 2017 and 2019.[5] Finding out the crucial mechanisms for flood tolerance in plants has become notably

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1 https://www.nrdc.org/stories/flooding-and-climate-change-everything-you-need-know.
2 https://www.fao.org/resources/digital-reports/disasters-in-agriculture/en/.
3 https://fdc.nal.usda.gov/fdc-app.html#/food-details/172420/nutrients.
4 https://hort.purdue.edu/newcrop/afcm/lentil.html.
5 https://www.downtoearth.org.in/news/climate-change/india-lost-crops-on-18-million-hectares-to-extreme-floods-from-2017-2019-govt-75506.

https://doi.org/10.1016/j.stress.2023.100130
Received 1 August 2022; Received in revised form 2 December 2022; Accepted 5 January 2023
Available online 6 January 2023
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**Table 1**
Primers for qPCR analysis.

| Gene name | Accession number | Forward primer | Reverse Primer |
|---|---|---|---|
| Aldolase | MF421715 (Yadav et al., 2020) | CTGCTGCTAATCAGGTGGAA | CTGGTGAGCCTAGAGGAGAATA |
| *accD acetyl-CoA carboxylase carboxyltransferase beta subunit* | YP_009141546 | GGGGGATATCGTTCTTACCG | AAAACTGACGCGGTACCATT |
| *petD-cytochrome b6/f complex subunit IV (petD)* | YP_009141566 | GGTTCTTTCGGAGGATGGAT | GATGGTTCGAGAACCGCTAA |
| *ribosomal protein L2 (rpl2)* | XM_004489738.2 | ATTGAGCTGAACCCAGGACAA | GACGTGGGCTCCTTCAAGATC |
| *Chloroplast Copper/Zinc Superoxide Dismutase* | MK605637.1 | ACCGCATTTTAATCCCAACA | ACCCTTTCCGAGGTCATCTT |
| *PBA Flash photosystem II D1 precursor (psbA)* | MH681286.1 | CCTATGGGGTCGCTTCTGTA | TTGCGGTCAATAAGGTAGGG |
| 18s *rRNA* | M16859.1 | CTGGTGGAATGCTCTCGAAAC | TGCGGATTTCACCCCTACAC |
| *HSP70* | NM_001357341.1 | GAAGATGGTTGAGGAAGCAGAGA | CCAACGCATTCTTTGCATCA |

relevant, especially with the rise in flooding. Flooding affects plants by creating anaerobic conditions for the plant, which slows respiration to the point that fermentation is the only viable pathway for energy (Wiebold, 2007). However, this pathway is inefficient for energy production and thus would lead to the death of the plant. Our research project aims to understand the molecular and physiological responses in *L. culinaris* under flood stress. In accomplishing this goal, our research aims to understand gene expressions of lentils under flood conditions at the physiological level using a variety of physiological tests as well as on a molecular level using qPCR to study the gene expression. Thus, we hypothesize that flooded lentils will exhibit differential expression of genes accompanied by other physiological responses compared to non-flooded lentils.

## 2. Material and methods

### 2.1. Plant materials and flood stress

Six lentil seeds were germinated by placing the seeds between 2 and 3 layers of damp medical gauze in a petri dish for four days in the dark. After four days, the germinated seeds were potted into a soil mixture of 30% perlite and 70% potting soil. Then 4–5 Osmocote Smart-Release Plant Food (Scotts Company, USA) pellets were evenly spread across the top of the soil. The soil was watered with regular tap water. The plants were then placed in a growth chamber at 24 °C with an average light intensity of 945 lux. The light in the growth chamber was set to a 12-hour photoperiod. Lastly, the plants were watered every three days for seven weeks. After seven weeks, to mimic the flood, three plants were placed in plastic trays filled with water until the water level was 3–4 cm above the soil (Supplementary File. 1). The rest of the three lentils were used as control plants (no flooding). Both groups were allowed to develop under the same conditions in the growth chamber. After four days of flooding, the stressed lentils began showing drooping and yellowing signs. Another set of seven lentil plants was grown under the same conditions (flooded: four plants and control: three plants) for physiological experiments and similarly stressed.

### 2.2. Gene expression studies

Total RNA was extracted on the fourth day from 100 mg of leaves per plant from three control (referred to here as C1, C2, and C3) and three stressed plants (referred to here as S1, S2, and S3) using the Trizol reagent (Thermofisher Scientific, USA) following Ramadoss and Basu, 2018. Once the RNA was extracted and purified, cDNA was synthesized using the iScript cDNA synthesis kit (Bio-Rad, USA) following the manufacturer's protocol, with 20 µl of total final reaction mixture with 1 µg of RNA, 4 µl of 5x iScript reaction mix and 1 µl of iScript Reverse Transcriptase. The reaction mixture was incubated in a thermocycler with the following settings - Priming at 25 °C for 5 mins, Reverse transcription (RT) for 20 min at 46 °C, and RT inactivation for 1 min at 95 °C and an additional step of hold at 4 °C. The gene expression was quantified by performing real-time quantitative PCR using iTaq Universal SYBR Green Supermix (Bio-Rad, USA). The genes that we tested for differential expressions were: *accD-acetyl-CoA carboxylase carboxyltransferase beta subunit (accD), aldolase, petD-cytochrome b6/f complex subunit IV (petD), photosystem II protein D1 precursor (psbA)*, and *chloroplast copper/zinc superoxide dismutase (SD)* (Table 1). Additionally, the *ribosomal protein L2 (rpl2)* (Jin et al., 2019), *18S rRNA* (Yu et al., 2019), and *heat shock protein 70 (Hsp70)* (Tang et al., 2016) genes were used as the reference genes. A Bio-Rad C1000 Touch(™) Thermal Cycler machine (Bio-Rad, USA) was used for qPCR. For the qPCR, the reaction master mix was prepared using the iQ™SYBR®Green Supermix kit (BioRad, USA). The 10 µl of the final reaction mixture is prepared as follows: 5 µl of iQ™SYBR®Green Supermix (2x), 1 µl of 10 µM forward primer, 1 µl of 10 µM reverse primer, 1 µl of the template cDNA (synthesized as described above) and 2µl of nuclease-free water. The setting for the qPCR was set to initial denaturation at 95 °C for 30 s once; then 39 cycles of denaturation at 95 °C for 5 s, annealing at 60 °C for 30 s; a melt curve from 65 °C to 95 °C with 5 °C incremental increases for 5 s; and a final plate read. Lastly, differential gene expression was viewed by a graph generated from the Bio-Rad C1000 Touch(™) Thermal Cycler machine (Bio-Rad, USA).

### 2.3. Physiological and biochemical studies

Another batch of seven (three control and four stressed) seven-eight weeks old plants, as mentioned before, were used for physiological studies consisting of chlorophyll content, catalase activity, sugar content, and moisture content estimations.

#### 2.3.1. Chlorophyll content

Chlorophyll content was estimated using the methods established by Hiscox and Israelstom (1979). Freshly collected leaves from each plant (0.5 g) were cut into small pieces, placed in test tubes with 5 mL of DMSO, and then heated in a water bath for 2.5 h at 50 °C. The absorbance of each bright yellow or green supernatant was measured at 645 nm and 663 nm and blanked using DMSO on a NanoDrop UV Visible Spectrophotometer (Thermo Scientific, USA). Lastly, total chlorophyll content was calculated based on the formula developed by Arnon et al. (1949). These included:

- Chlorophyll a (mg/ mL) = $(12.7 \times Abs_{663\ nm})$ - $(2.69 \times Abs_{645\ nm})$
- Chlorophyll *b* (mg/ mL) = $(22.9 \times Abs_{645\ nm})$ - $(4.68 \times Abs_{663\ nm})$
- Total chlorophyll (mg/ mL) = chlorophyll *a* + chlorophyll *b* (S.E.R.A.S, 1994)

#### 2.3.2. Catalase activity

Freshly collected leaves from three control plants (4.0 g total from three control plants) and three flood-stressed plants (4.0 g total from three stressed plants) were ground in 8 mL of distilled water separately. Then 1.6 mL of the aqueous solution was pipetted into a 50 mL graduated cylinder, and 1.6 mL of 3% $H_2O_2$ was added. Finally, catalase activity was measured by the volume of effervescence as described by Euler and Josephson (1927).

#### 2.3.3. Sugar content

Sugar content was estimated using the methods established by Hodge and Hofreite (1962) and Sadasivam and Manickam (1992). Fresh

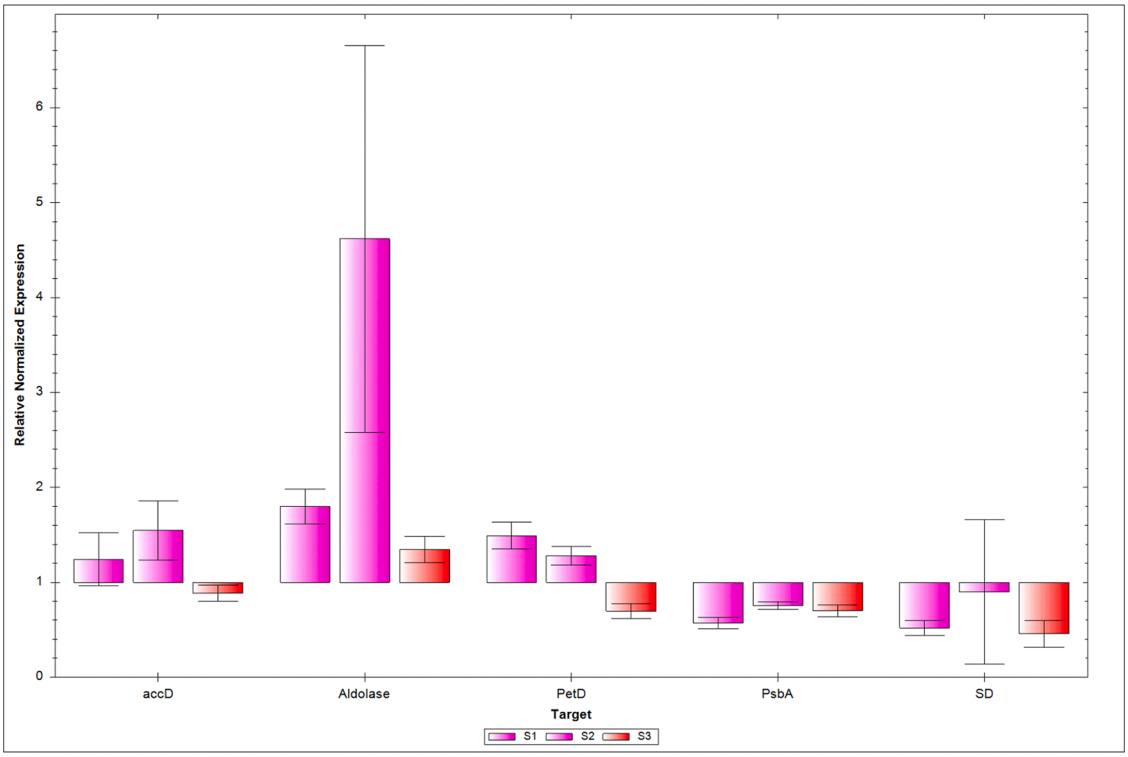



**Fig. 1.** qPCR data of the gene expression of *acetyl-CoA carboxylase carboxyltransferase beta subunit (accD), aldolase, petD-cytochrome b6/f complex subunit IV (petD), PBA flash photosystem II D1 precursor (psbA)*, and *chloroplast copper/zinc superoxide dismutase (SD)* in flooded lentils normalized against the non-flooded lentils' gene expressions. Additionally, *18S rRNA, heat shock protein 70 (Hsp70)*, and the *ribosomal protein L2 (rpl2)* were used as reference genes. The ct (cycle threshold) values of three non-flooded lentil plants were averaged and used as 'control' values for normalization against the stressed plants. S1, S2, and S3 = three individual flood-stressed plants. Each bar represents one biological replicate. In addition, each bar represents the mean of 3 technical replicates.

leaves from each plant (0.1 g) were placed in test tubes with 5 mL of 100% HCl, heated for three hours at 50 °C in a water bath, and then cooled to room temperature. The solutions were neutralized to a pH of about 7 by adding solid sodium bicarbonate. Then each solution was brought up to a volume of 50 mL using distilled water in 50 mL tubes. These tubes were then centrifuged at 3900 rpm for 5 min at 24 °C. Once centrifuged, 0.4 mL of the aliquot was added to test tubes with 4 mL of Anthrone reagent (200 mg of Anthrone was dissolved in 100 mL of ice-cold 95% of sulfuric acid ($H_2SO_4$)). The test tubes were heated for 8 min at 95 °C and then cooled on water and ice. The absorbance of each yellow supernatant was measured at 630 nm and blanked using the Anthrone reagent on a NanoDrop UV Visible Spectrophotometer.

#### 2.3.4. Relative moisture content (RMC)

Relative moisture content was estimated using 10 leaves from each plant. These leaves were weighed before and after being placed in distilled water for four hours. Then they were dried for 14 h in an incubator set to 43 °C-45 °C and weighed again. Lastly, moisture content percentage was calculated as RMC (Singh and Rao, 1983):

$$(\mathbf{Freshweight} - \mathbf{Dryweight})/(\mathbf{Submergedweight} - \mathbf{Dryweight})$$

### 2.4. Microscopy

Leaves and stems of control and flood-stressed plants fixed in 70% ethanol were washed a few times in distilled water and stained overnight with 0.1% aqueous Safranin 'O'. Excess leaf and stem samples were stored in 70% ethanol at 4 °C for later use. Images were captured using VK-X3000 series hybrid optical 3-D surface profiler 3100 version laser scanner confocal microscope (Keyence Corporation of America, USA) to gather data on surface texture scans and analyze the topography of epidermal wax on leaves and stems of control and flood stressed plants.

### 2.5. Volatile studies

Volatile emission analyses were conducted on three flood-stressed and three control plants. The plants were housed in a temperature-controlled (25 °C) environmental chamber at the University of California, Irvine. At the end of the 4-day flood stress, three sets of leaves were excised from each plant. Each set of leaves was placed in a glass vial with a screw cap and septum. The vial was then positioned underneath a high-intensity lamp for 15 min to allow the volatile emissions from the excised leaves to accumulate in the vial's headspace. 0.5 mL of diluted gas-phase tetramethylethylene (TME) was subsequently injected into each vial to serve as an internal standard. The vial was then placed in a headspace sampling system (HSS) (INFICON Inc., East Syracuse, NY, USA) and heated to 50 °C. The volatiles in the vial were then transferred to a portable gas chromatograph-mass spectrometer (GC–MS) (HAPSITE ER; INFICON Inc.) via a heated transfer line (60 °C). The volatiles were first pre-concentrated onto a triple-bed carbon trap. Then, the trap was desorbed, and the volatile compounds were separated using a DB-1 chromatographic column (length: 15 m; internal diameter: 0.25 mm; film thickness: 1.0 μm). Nitrogen was used as the GC carrier gas. The GC oven temperature program used during the analyses was as follows: initial temperature, 60 °C; 1 min hold; Ramp 15 °C/min to 90 °C; Ramp 8 °C/min to 170 °C; Ramp 15 °C/min to 200 °C; and 1 min hold (total GC runtime was 16 min). The MS was operated in "full scan" mode, and we scanned for mass fragments in the range of $m/z$ 39 to 239. The chromatographic peaks of each detected compound were integrated using the Smart IQ software (version 1.30; INFICON Inc.). The peak areas of the compounds of interest were normalized to the peak areas of the internal standard and to the fresh weight of the leaves.

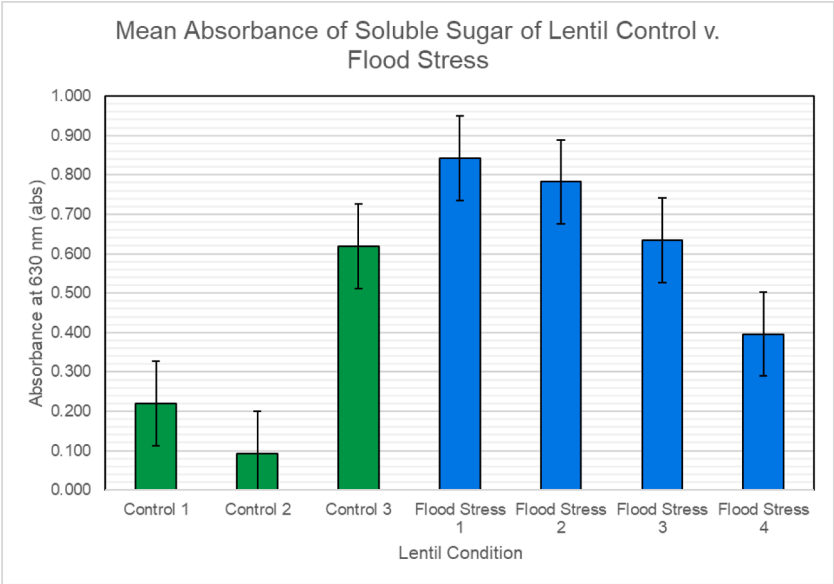



**Fig. 2.** Average absorbance of 3 technical replicates for each of the biological replicates under control conditions ($n = 3$), or under flooded conditions ($n = 4$). The bars placed above each of the graphical bars represent standard error. Welch's *t*-test: *t*-value: −2.19 and *p*-value: 0.04.

### 2.6. Statistical analysis

Each experiment consisted of at least 3 replicates. However, for catalase activity, only one replication per experiment was used. MS Excel was used to calculate the mean and standard error. Welch's *t*-test was performed using R and Rstudio (4.2.0) to analyze the significant differences between control and flood-stressed lentil plants (Supplementary file 2).

## 3. Results and discussion

### 3.1. Molecular studies

Under hypoxic conditions, Subbaiah and Sachs (2003) found that there is a drastic change in the synthesis of anaerobic proteins (Subbaiah and Sachs, 2003). Most of these anaerobic proteins are important enzymes for either glycolysis or sugar-phosphate metabolism (Subbaiah and Sachs, 2003).

Aldolase is an important enzyme for sugar-phosphate metabolism and an anaerobic protein (Subbaiah and Sachs, 2003). The aldolase gene is an enzyme responsible for the catabolism of fructose-1,6- bisphosphate into glyceraldehyde-3-phosphate and dihydroxyacetone phosphate in processes like glycolysis (Lv et al., 2017). In our case, two of the flooded lentil plants (out of three) had an increase in the expression of this gene in contrast to the control plants (Fig. 1), similar to other published research (Subbaiah and Sachs, 2003). It was found that anaerobic conditions due to flooding in maize seedlings resulted in a 70% increase in the translation of the aldolase protein (Subbaiah and Sachs, 2003). Furthermore, upregulation of the aldolase gene was also observed in flooded *Arabidopsis thaliana* and soybean (*Glycine* max) plants since they had increased levels of aldolase proteins over the non-flooded plants (Khan et al., 2022). These higher levels of aldolase gene expression are due to an increased need for glycolysis under abiotic stresses like flooding. Glycolysis is needed as a substitute to support the energy demands of those plants under flooding since oxidative phosphorylation is no longer a viable pathway for energy (Khan et al., 2022). Additionally, upregulation of this gene is a result of the increase in the amount of soluble sugar produced within the flooded lentil (Fig. 2). The third stressed plant (S3 in Fig. 1) did have a slight downregulation of the gene, probably due to senescence at the cellular level in the S3 plant.

*Acetyl-CoA carboxylase carboxyl transferase beta subunit (accD)* gene helps in fatty acid biosynthesis in plastids (Nakkaew et al., 2008). Fatty acid biosynthesis involves multiple reactions, and it begins with the carboxylation of acetyl-CoA to malonyl-CoA with the help of the acetyl-CoA carboxylase complex.[6] Previous reports have suggested that plants can increase their resistance to drought stress by inhibiting the production of acetyl-CoA carboxylase, which ultimately reduces the synthesis of fatty acids (Gu et al., 2020). However, we were unable to find any previous studies on increased *accD* expression under flooded conditions. Proteomic and lipidomic analysis of *Arabidopsis thaliana* plants under hypoxia exhibited higher levels of lipids (Xie et al., 2021). Thus, we can assume that the expression of the *accD* increases with increased fatty acid levels. Additionally, Madoka et al. (2002) reported that overexpression of *accD* in tobacco promoted leaf longevity. Herein our study, the beta subunit of acetyl carboxylase was upregulated in two (S1 and S2 in Fig. 1) out of three plants under flood stress. Hence, we can hypothesize that plants can improve their flood stress by upregulating the accumulation of accD, which in turn improves leaf longevity.

Another gene known as *petD (petD-cytochrome b6/f complex subunit IV)* transcribes a protein of $b_6f$ cytochrome complex,[7] whose expression appeared to be upregulated in two out of three flood-stressed plants (S1 and S2 in Fig. 1) in our experiment. This protein plays an important role in electron transfer during photosynthesis. In general, flooding decreases overall net photosynthesis (Liu et al., 2013). We predict that electron transport was either not affected or slightly increased post-flooding, as previously reported by Doffo et al. (2018). Interestingly, although net chlorophyll content was reduced following flooding, the overall production of electron transport enzyme cytochrome b6/f complex was not hampered in two out of three plants (Fig. 1). Providing a plausible explanation for this observation will require further research.

The *chloroplast copper/zinc superoxide dismutase (SD)* gene is responsible for the production of superoxide dismutase (SOD). These SODs are enzymes that catalyze reactive oxygen species (ROS), like superoxide ($O_2^-$), into hydrogen peroxide ($H_2O_2$) (Wang et al., 2018). Under flood conditions, all three lentil plants in our experiment displayed downregulation of this gene compared to the control plants (Fig. 1). Flooding causes inhibition of photosynthetic systems, which results in higher levels of superoxide (Hossain et al., 2009). Thus, we

[6] https://www.ebi.ac.uk/interpro/entry/InterPro/IPR000438/.
[7] https://www.uniprot.org/uniprotkb/A0A023IN75/entry.

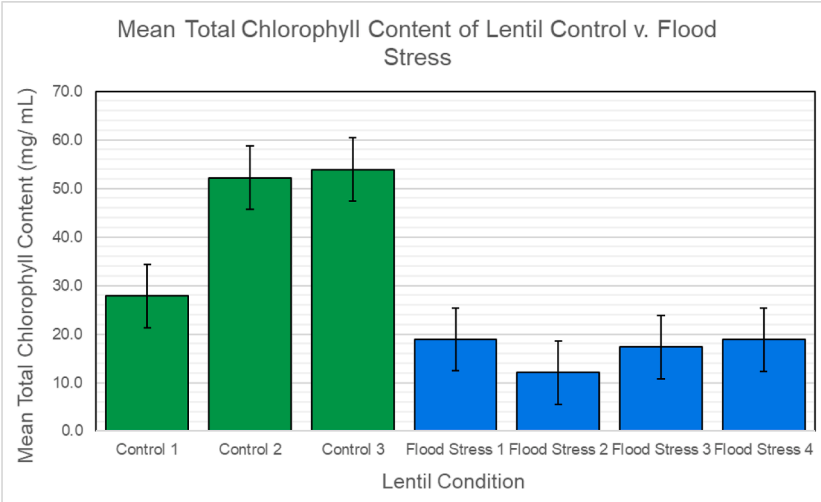



**Fig. 3.** The mean of 3 technical replicates in each biological replicate of the total chlorophyll content present. Control: $n = 3$, and Flood: $n = 4$. The bars above the graphical bars represent standard error. Welch's *t*-test: *t*-value: 5.8 and *p*-value: 1.298e−05.

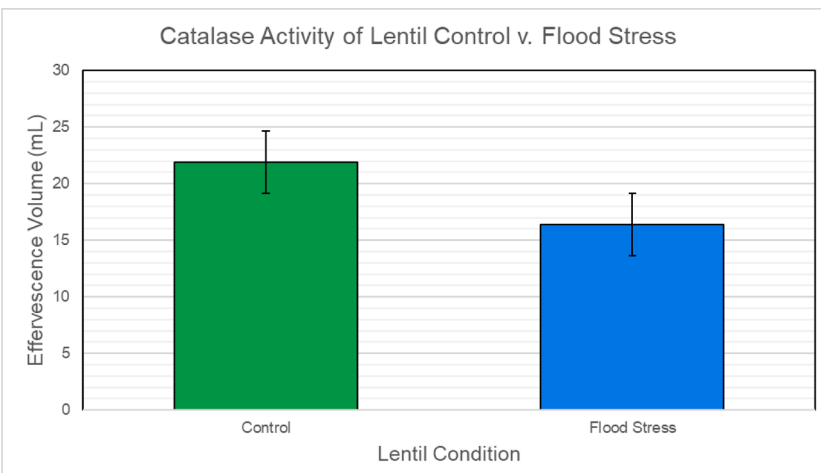



**Fig. 4.** The catalase activity is represented as the mean of the 3 biological replicates ($n = 3$) for control, and the mean of the 4 biological replicates ($n = 4$) under flooded conditions. The bars placed above each of the graphical bars represent standard error.

expected higher expression of SOD's enzymes to catalyze the influx of superoxide. To account for this contradiction, we speculate that the downregulation of SOD's is likely due to the senescence of the lentil plant leaves. This downregulation or reduction in SOD activity was also seen in the senescing tobacco leaves subjugated to flooding (Hurng and Kao, 1994). In all likelihood, the reduction in SOD is correlated with the general decrease in proteins within the cells (Dhindsa et al., 1981).

*PBA flash photosystem II D1 precursor (psbA)* gene encodes the D1 protein, which comprises part of the reaction center for Photosystem II (PSII).[8] PSII is responsible for the abstraction of electrons from water using light energy in order to generate a proton gradient and oxygen[7]. Previous research has established that flooding reduced the expression of the *psbA* gene inside water-stressed wheat cultivars (Liu et al., 2006). Similarly, we observed a decreased expression of the *psbA* gene within flooded lentils (Fig. 1). We hypothesize that the downregulation of *psbA* eventually affected PSII. As mentioned before, flooding reduces photosynthetic activity. Although the mechanism for the reduction in *psbA* gene expression and PSII activity is not well understood, it may be due to reduced RNA stability under flood stress (Liu et al., 2006). However, there may be a possibility to increase flood tolerance by overexpression of the *psbA* gene. Data from a study on drought tolerance in tobacco showed that the overexpression of the *psbA* gene increased the amount of D1 proteins and improved drought tolerance (Huo et al., 2016). This may be a possible avenue to explore in increasing the tolerance to flooding in lentils and would require further testing.

### 3.2. Physiological studies

To detect the influences of hypoxia and ROS species, we measured sugar content, chlorophyll content, catalase activities, and moisture content,

#### 3.2.1. Soluble sugar content analysis

It was observed that there were elevated levels of absorbance for soluble sugars in flood-stressed lentils compared to non-flooded lentils (Fig. 2). This effect has also been noted in other studies. For instance, flooding resulted in higher levels of soluble sugars in the roots and shoots of maize seedlings (Pourabdal et al., 2008). Moreover, an increase in soluble sugar in the leaves and stems of flooded tomato plants was also observed (Mignolli et al., 2021). Under hypoxic conditions, plants switch to using the fermentation pathway as a substitute for their

[8] https://www.uniprot.org/uniprotkb/A0A023IN49/entry.

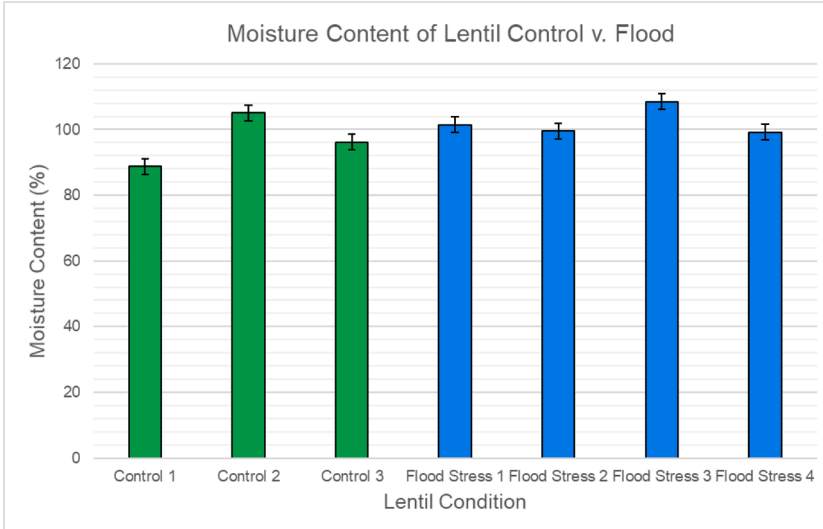



**Fig. 5.** Relative moisture content in the leaves of each biological replicate under flood conditions ($n = 4$), and for each biological replicate under control conditions ($n = 3$). The bars placed above each of the graphical bars represent standard error. Welch's *t*-test: *t*-value: −1.16 and *p*-value: 0.29.

energy source. For fermentation to occur, it requires soluble sugars as a substrate for subsequent metabolic processes (Miro and Ismail, 2013). Therefore this allows the plant, lentils in our case, to endure flooding for a short period of time.

### 3.2.2. Chlorophyll content analysis

To understand the physiological responses of plants under stressed conditions, analysis of green pigment compositions can give informative results (Amri et al., 2014). In our studies, we have seen a decrease in chlorophyll content in the leaves of flood-stressed plants (Fig. 3). Previous reports have shown that root cell death can cause a reduction in chlorophyll content with a dry and yellowish appearance of leaves (Amri et al., 2014). Root cell death can be induced by water stress which impacts the root functionality, such as water or mineral uptake (Amri et al., 2014).

### 3.2.3. Catalase content analysis

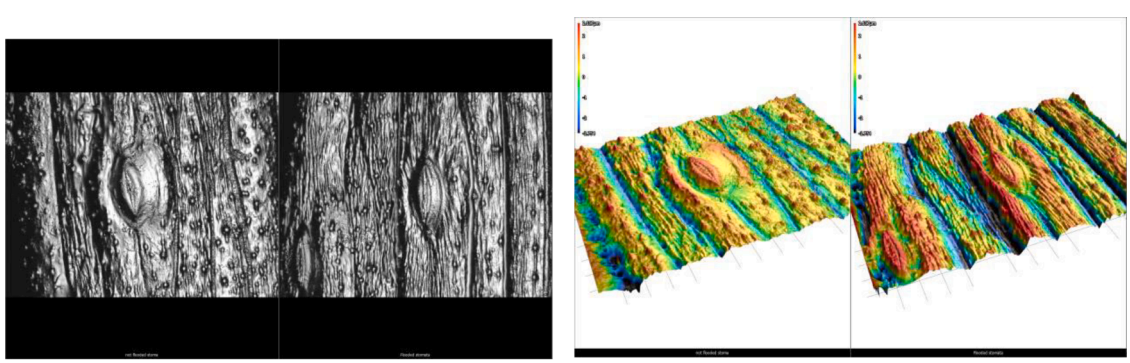
The catalase enzyme responsible for decomposing ROS into water and oxygen is known as hydrogen peroxide (Wang et al., 2018). In our experiment, it was observed that catalase activity was lower in flood-stressed lentils as compared to the control lentils (Fig. 4). Results for catalase activity vary by researchers. It was noted that catalase had no apparent change in senescing leaves of tobacco plants (Hurng and Kao, 1994). However, catalase activity was observed to have increased on the first day of flooding or during a short duration of flooding (Yan et al., al.,1996 and Park and Lee, 2019). Catalase activity can decrease as the duration of flooding increases (Yan et al., 1996). We theorize that senescence in leaves exhibits lower catalase activity. This is likely due to the downregulation of the superoxide dismutase gene. Prolonged flooding also decreased superoxide dismutase activity (Yan et al., 1996), ultimately decreasing catalase activity.

### 3.2.4. Relative moisture content analysis

We measured the relative moisture content and found that submerged weight (as described in the RMC protocol before) did not change compared to fresh weight. However, dry weight has changed drastically for all controlled and stressed plants. Surprisingly, moisture content has not shown any significant difference between controlled and flood-stressed plants (Fig. 5).

## 3.3. Leaf and stem surface studies

Under different biotic and abiotic stress conditions, plants show some common traits that can be considered adaptive (Tellechea-Robles et al., 2019). In the case of the flood stress condition, one of the common responses of plants is stomatal closure in order to lower the $CO_2$ uptake, which can be seen in our observations as well (Rodríguez-Gamir et al., 2011) (Fig. 6A and 6B). The flooded lentils in this study also had



(A) (B)

**Fig. 6.** **(A)** Surface scan of the stomata of lentils. Left = Not Flooded and Right = Flooded. **(B)** The figure presented here displays the stomatal 3D wax scan of lentils. Left = Not Flooded & Right = Flooded. Redness is representative of wax.

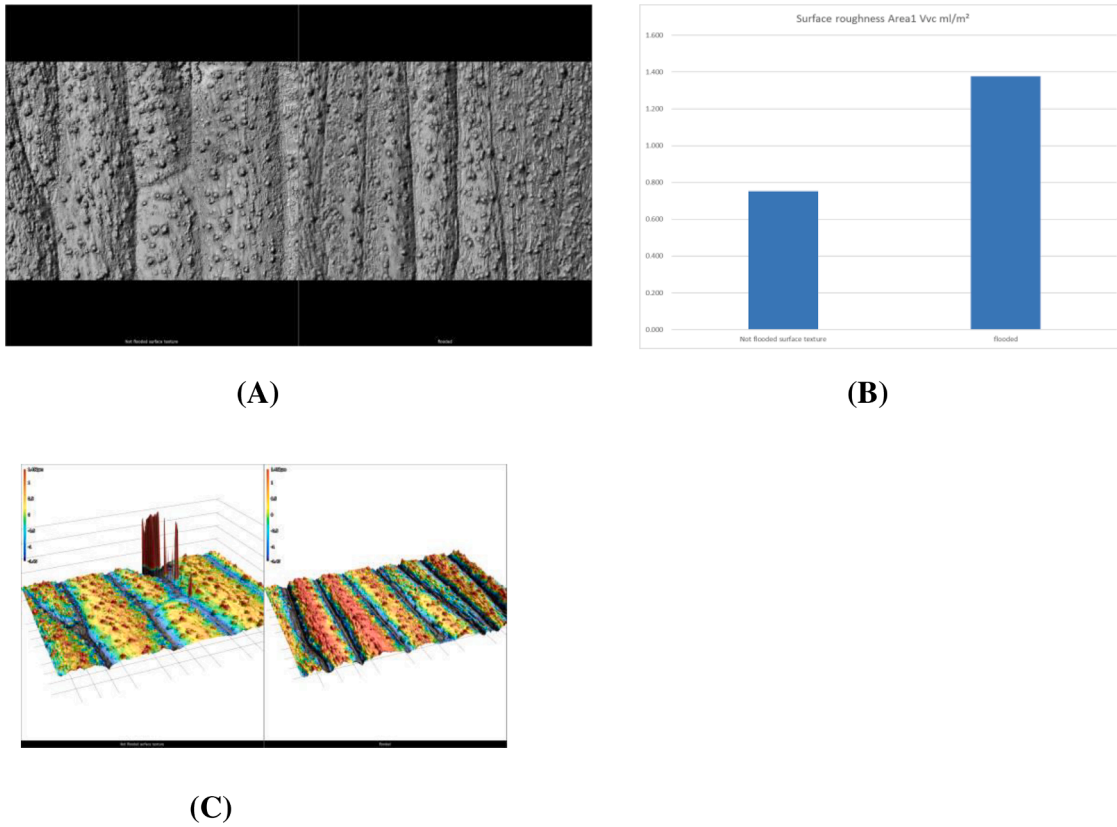



**Fig. 7.** **(A)** Surface texture scan of the stem of lentils. Left = Not Flooded and Right = Flooded. **(B)** This graph describes the surface texture of the stem. Left = Not Flooded and Right = Flooded. **(C)** The figure shown here represents the 3D wax scan of lentil stems. Left = Not Flooded and Right = Flooded. Redness is representative of wax.

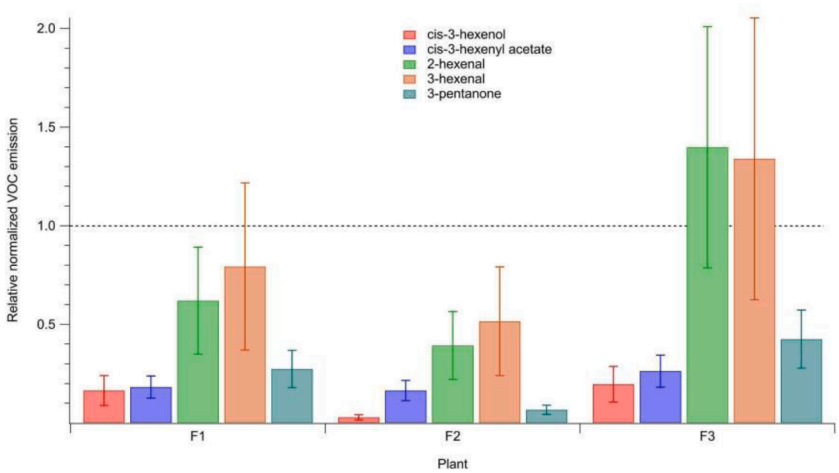



**Fig. 8.** Normalized GC–MS Volatile Compound Analysis. The volatiles measured were cis-3-hexenol, cis-3-hexenyl acetate, 2-hexenal, 3-hexenal, and 3-pentanone. Volatiles measured in flooded lentils were normalized against the control lentils. F1, F2, and F3 represent the 3 flooded lentil plants. Values that are less than 1, denote the down-regulation of emissions in flooded lentils compared to the control. However, values that are greater than 1, denote the up-regulation of emissions in flooded lentils compared to the control.

elevated levels of surface roughness compared to the non-flooded plants (Fig. 7A and 7B). Another adaptive trait is cuticular waxes (Tellechea-Robles et al., 2019). It has been reported that cuticular waxes play a vital role in the abiotic stress-related responses in plants (Dhanyalakshmi et al., 2019). In our research, we also found higher wax accumulation in flood-stressed lentil plants (Figs. 6B and 7C). Wax accumulation has the potential to cope with crop plants under flood stress (Dhanyalakshmi et al., 2019). Previous studies found that leaf wettability (hydrophilic affections of leaves) can be determined by epicuticular wax formation (Tellechea-Robles et al., 2019). The higher wax formation gives a higher leaf hydrophobicity, which is reported in flood-stressed plants (Tellechea-Robles et al., 2019). Hydrophobicity is further improved with increased surface roughness (Tellechea-Robles et al., 2019).

### 3.4. Volatile studies

For the volatile studies, 5 compounds were identified, that is, cis-3-hexenol, cis-3-hexenyl acetate, 2-hexenal, 3-hexenal, and 3-pentanone. Most of these compounds (i.e., all except for 3-pentanone) were from

the lipoxygenase pathway (LOX-pathway), which is responsible for signaling and fragrances emitted by plant cells. Emissions of products of the LOX pathway indicate stress (Kreuzwieser et al., 2013 and Niinemets et al., 2018). We note that the LOX-pathway emissions observed in our study could also have been induced by the measurement approach, where the leaves were excised from the plant prior to sampling. It has been reported that LOX emissions are higher for flood-sensitive plants than flood-tolerant plants (Kreuzwieser et al., 2013). In our study, the emissions were higher in control plants compared to flood-stressed plants (Fig. 8). Thus, we hypothesize that flood-stressed plants had reduced photosynthesis and had a lower amount of carbon substrates to make LOX.

## 4. Conclusion

From our studies, we have concluded that lentil plants are highly flood-sensitive plants. Under flooded conditions, they responded to hypoxia by altering various biochemical pathways for the first two to three days. However, after four days, the plants showed senescence and visible signs of injury due to flood stress. It is evident that the cells differentially express genes and produce volatiles to cope with flood stress. It should be noted here that gene expression data obtained through qPCR also depend on the physiological status of the plants. Also, not all reference genes are suitable for studying all abiotic stress-induced gene expressions (Ma et al., 2013). Understanding the molecular physiology of lentils will allow us to develop flood-tolerant lentils in the future. It will also help us to understand how the plant will perform in the changing environment.

## Declaration of Competing Interest

The corresponding author, on behalf of all the authors of a submission, declare no competing interest.

## Data availability

Data will be made available on request.

## Acknowledgments

We sincerely thank Dr. Thomas Terrill, Fort Valley State University, GA, USA, for critically reviewing the manuscript.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.stress.2023.100130.

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